B2.1 Membranes and membrane transport

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b2-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 223–236 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

The bilayer

B2.1.1 Phospholipids form bilayers in water

pp. 223–224

A phospholipid has a **hydrophilic** phosphate head and two **hydrophobic** hydrocarbon tails, so it is **amphipathic**. In water, the heads face the water and the tails face one another. This forms a continuous double layer, the basis of every cell membrane.

B2.1.2 The bilayer acts as a barrier

p. 224

The hydrophobic core has low permeability to large molecules and hydrophilic particles, including ions and polar molecules. The membrane therefore separates two watery solutions.

Passive movement

B2.1.3 Simple diffusion

pp. 225–226

Diffusion is the net movement of particles from higher to lower concentration, and it uses no ATP. Oxygen and carbon dioxide are small and non-polar, so they pass between the phospholipids.

B2.1.4 Integral and peripheral proteins

pp. 226–227

Integral proteins are embedded in one or both lipid layers, and the parts inside the core are hydrophobic. **Peripheral proteins** are attached to one surface. Membrane proteins vary widely in their structure and role, and more active membranes contain more of them.

B2.1.5 Osmosis and aquaporins

pp. 227–229

Osmosis is the net movement of water across a membrane, from a lower to a higher solute concentration. Water molecules move at random, and the membrane does not let the solute cross. **Aquaporins** are channel proteins that greatly speed up water movement while blocking ions.

Watch out: describe direction using solute concentration. Water moves, and the solute does not.

B2.1.6 Channel proteins and facilitated diffusion

p. 229

A channel has a hydrophilic pore, and its size and charge decide which particles pass. Many channels open and close. Movement is passive and goes down the concentration gradient.

Active movement and selectivity

B2.1.7 Pump proteins and active transport

pp. 230–231

Pumps use **energy from ATP** to move specific particles **against** a concentration gradient. ATP makes the protein change shape, and pumps usually work in one direction.

Route	ATP?	Protein	Direction
Simple diffusion	No	None	Down gradient
Facilitated diffusion	No	Channel	Down gradient
Active transport	Yes	Pump	Against gradient

B2.1.8 What makes a membrane selective

p. 232

Facilitated diffusion and active transport are selective, because each protein handles particular particles. Simple diffusion is not selective, since it depends only on size and on hydrophobic or hydrophilic properties.

The whole membrane

B2.1.9 Glycoproteins and glycolipids

p. 234

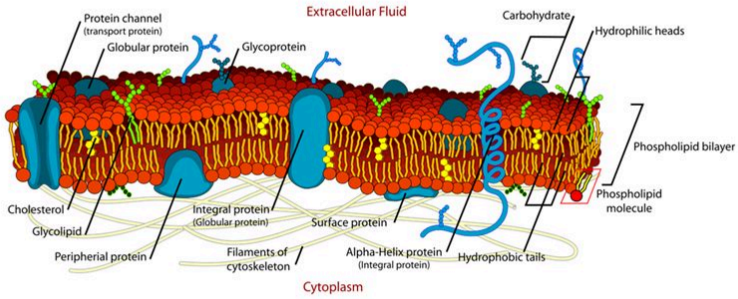
These have a carbohydrate attached to a protein or a lipid, and the carbohydrate is always on the extracellular side. They help with **cell adhesion** and **cell recognition**.

B2.1.10 The fluid mosaic model

pp. 234–236

The fluid mosaic model

B2.1.10



Fluid: phospholipids and many proteins move sideways freely.

Mosaic: proteins are scattered through the bilayer like tiles.

Be ready to draw it, labelling the bilayer, heads, tails, integral and peripheral proteins, glycoproteins, glycolipids and cholesterol.

Cell membrane. LadyofHats, public domain.

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Molecules move sideways within the membrane, which makes it fluid, and proteins are scattered among the lipids like a mosaic. A drawing should show phospholipids with hydrophilic and hydrophobic regions, integral and peripheral proteins, glycoproteins and cholesterol.

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